AKSHAYMANI SORNALINGAM

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EDUCATION

University of Southern California **Los Angeles, CA**
Master of Science in Healthcare Data Science; *GPA: 3.75/4.0* **AUG 2023-MAY 2025**
Awards: Viterbi School of Engineering Dean's Merit Scholarship- Fall 2023, Sol Price School Scholarship: 2023-2024
Relevant Coursework: Machine Learning, Data Analytics with Python, Statistics, Databases, Experimental Design

Anna University (AU-AC Tech) **Chennai, India**
Bachelor of Technology in Industrial Biotechnology; *GPA: 3.97/4.0 (9.5/10)* **JUL 2019-MAY 2023**

SKILLS

Languages/Databases: Python, R, SQL (MySQL), MongoDB
Analytics: Pandas, Numpy, Matplotlib, Seaborn, Tableau, PowerBI, Beautiful Soup, Selenium, Excel, Minitab, Hypothesis Testing
Cloud & Data Engineering: Snowflake, DBT, AWS (EC2, S3), Apache Airflow, ETL
AI Specific Techniques: Scikit-learn, Tensorflow, Keras, PyTorch, OpenCV, SciPy, Langchain

WORK EXPERIENCE

Data Scientist, Farmers Insurance, Woodland Hills, CA **MAY 2025-PRESENT**
- Involved in the development and deployment of **Unconstrained Pure Pricing models for Auto Insurance**
- Orchestrated an extraction pipeline to parse over **5 million raw JSON payloads** obtained using a **snowflake connector**
- Converted the payload into a **4-feature clean vector store dictionary** and enriched it with **5+ aggregated features**

Data Analyst Intern, Data-sleek, Los Angeles, CA **JAN 2025-MAY 2025**
- Conducted customer analysis of **120 Clients**, assessing profitability metrics using **Pandas & Advanced Excel**
- Explored AI applications for Healthcare & Finance industry seeking to increase their business efficiency by **~30%**
- Used **Snowflake-schema tables** to analyze **100,000+ clockify entries,** enabling financial analysis for Data-Sleek
- Built **4 interactive Story Dashboards** in **Tableau** to visualize billable vs Non billable hours from 2020 to 2024

Graduate Student Researcher, USC Radiomics Lab, Los Angeles, CA **FEB 2025- MAY 2025**
- Implemented an **AI-driven software pipeline for renal tumor mass classification** by combining imaging and clinical data
- Optimized data ingestion pipelines to extract clinical data from **EHRs of FHIR Standard** using RESTful API endpoints
- Developed a **MLP** with **0.91 AUC** in classifying renal tumors using PCA reduced image features and clinical data
- Transferred clinical data into **AWS RDS**, improving SQL query performance by **40%** using ORM-based interactions
- Expanded software capabilities using **Django** for backend integration and enhanced UI interactivity with **Vue**
- Deployed the entire pipeline in a multi container **Amazon EC2 Linux** instance via **Amazon Elastic Container Registry**

Graduate Research Assistant, USC Dornsife, Los Angeles, CA **AUG 2024-NOV 2024**
- Cumulated information about basic parameters and benchmarking information of **300+** RNA seq tools using **Selenium**
- Achieved a **Data coverage of 89%** in data collection process with an overall **data accessibility** rate of **94.6%**
- Enhanced the Method Section of research **by 100%** by retrieving Git Parameters of **152 tools** using **REST API Git Calls**

Quality Assurance Data Analyst Intern, Alembic Pharmaceuticals Limited, Gujarat **JUL 2022-AUG 2022**
- Recorded Production QA parameters of **500 Oral Solid Dosage forms** as an Annual product Quality Review (APQR) report
- Checked if parameters were within USFDA standard limits by utilizing **control charts** with **Minitab statistical software**
- Performed Batch-wise EDA for 500+ tablets showing a **35% increase in production stability** in dry compaction stage
- Interpreted QC data to identify trends using **SPC**, improving decision making processes in inventory control by **40%**

PROJECTS

Determining the likelihood of Diabetes based on food habits **JAN 2024-MAY 2024**
- Produced **k-means clusters** with a **Calinski-Harabasz** Index value of **340.9** and a **Davies-Bouldin** Index value of **0.69**
- Developed a **Lasso regularized Logistic regression pipeline** using **Scikit-Learn** with a **Pseudo R square value of 0.96**
- **Correlated** dietary levels of **100+ individuals** and output variable transitively with Clinical results of disease predictors
- Defined an **SVM model** to classify user input consumption levels as diabetes positive or negative with an **accuracy of 0.91**

Nutritional data analysis of fast-food outlets **NOV 2023-DEC 2023**
- Scrapped Nutritional data of **200 food items** from KFC and Chick-Fil-A static webpages with **BeautifulSoup**
- Integrated the dataset in **MySQL Database** and manipulated data points in python script using **mysql.connector** module
- Leveraged **Pandas & NumPy** to preprocess data and conduct **statistical analysis** on nutrient trends between the two outlets